Do Now · retrieval

Lesson 066 · Problem Decomposition: Build a Useful Solution

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What makes a problem suitable for computation?

Q2

What is an abstraction?

Q3

What is a procedure?

Q4

Stack initially empty: Push A, B, C; Pop; Push D. Give the stack top-to-bottom.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Define and predict — Decomposition divides a larger problem into manageable sub-problems whose inputs and outputs are clear.

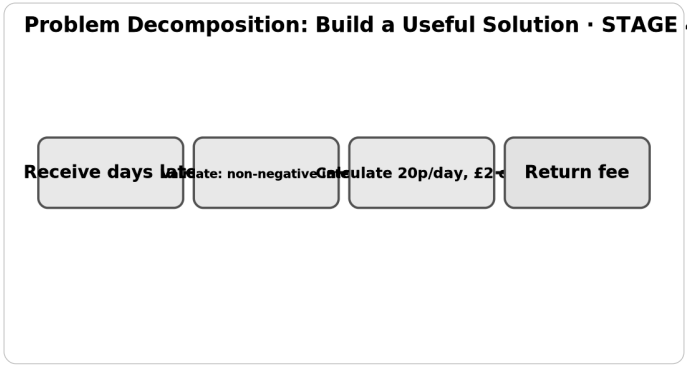
Core idea 2

Explain — A top-down design begins with a main task, then refines it into independently testable functions or procedures.

Core idea 3

Evaluate — Interfaces between subproblems matter: validation should happen before data are stored or a fee is charged.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

decomposition — Split task into smaller problems.

top-down design — Start broad, refine into subtasks.

interface — Inputs/outputs between components.

modularity — Small cohesive program components.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

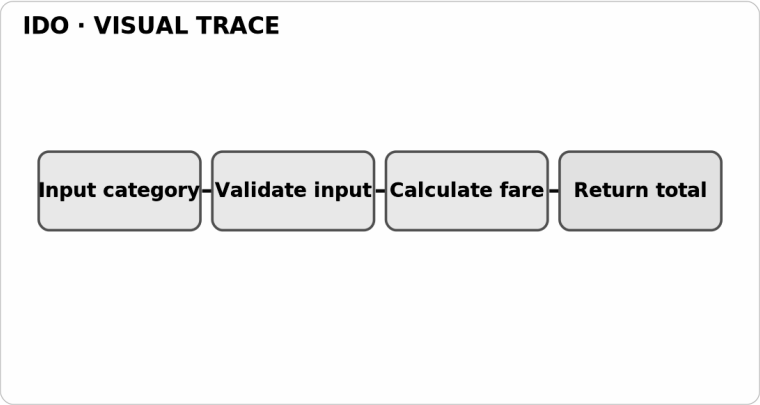
Problem

Decompose a cinema price calculator for child/adult tickets.

Follow the worked method

1. Follow the reasoning — State the interface for each subtask; every step has a specific responsibility.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Follow the reasoning
Reservation must occur before confirming success; check for failures.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Identify three independent subtasks and their outputs for a library fine system. [6]

Question 2

Write C# Fine(daysLate) for 20p/day capped at £2, zero for on-time returns. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain two benefits and one limitation of decomposition when a team builds a school booking app. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Splitting work means the components never need to communicate.

Use key vocabulary from memory

1. Explain decomposition in your own words.

2. Explain top-down design in your own words.

3. Explain interface in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.